

Abhay Sharma

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☎ (437) 223-7480

SKILLS

Languages: Python, SQL, Terraform (HCL), R, JavaScript/TypeScript, C#, Java

AI/ML: LangChain, LangGraph, OpenCV, TensorFlow/Keras, Scikit-learn, PyTorch

Cloud & Data: Azure (ADLS, ADF, Databricks, AI Search, App Services, Key Vault, Blob Storage), AWS (Bedrock, Lambda, Glue, Redshift, SageMaker, S3), Jenkins, Airflow, Docker, Kubernetes, PySpark, PostgreSQL, Git, Linux

WORK EXPERIENCE

Generative AI / Platform Engineer

Nov 2025 – Present

Manulife Financial

Toronto, ON

- Enabled **40+** cross-business unit deployments for **4** GenAI applications (including AI Voice Coach) by productionizing Azure infrastructure using **Terraform**, **APIM**, **VNets**, Private Endpoints, and CI/CD (**Jenkins**, **GitHub Actions**).
- Built **Python** and **SQL** ingestion pipelines in **Databricks** and RAG workflows for **5** enterprise knowledge sources, improving retrieval performance through optimized indexing strategies.
- Developed a **LangChain**-based RAG knowledge agent integrating **ServiceNow**, **Confluence**, and **Jira** with **OpenAI** models via **AI Foundry** and **APIM** to streamline internal knowledge retrieval and documentation workflows.

Data Science Intern

May 2024 – Dec 2024

Innovation, Science and Economic Development Canada

Ottawa, ON

- Built a **LangGraph** RAG agent in **SageMaker** with **GPT-4** and a self-service chatbot integrating **Bedrock**, **S3**, **RDS**, **SharePoint**, **OpenSearch**, and **API Gateway**, reducing manual workload by **~30%** and support requests by **~25%**.
- Improved **Beneficial Ownership Registry (BOR)** retrieval accuracy by **~20%** by designing ETL pipelines and LLM-powered naming conventions using **Python**, **SQL**, and **AWS**.

Data Engineer Intern

May 2023 – Aug 2023

Manulife Financial

Toronto, ON

- Reduced ETL runtime by **50%** (**4h** → **2h**) by migrating **22 of 30** ServiceNow tables to Azure using **Databricks**, **PySpark**, **Blob Storage**, **Data Factory (JSON)**, and **Jenkins** while resolving curation-layer data issues.
- Automated migration of **450 database schemas** using **Shell scripts** and **SQL** integrity checks, minimizing downtime.

Data Science Engineer Intern

Sep 2022 – Dec 2022

Manulife Financial

Toronto, ON

- Engineered a cost-saving data pipeline by mapping Canada's Deceased List (CDL) data with annuities data into **Azure Data Lake Storage** using **Fuzzy Matching** and **PySpark**, reducing **annual costs** by **~\$400,000**.
- Researched **Azure ML**, **OpenCV**, and **Prodigy** image detection models for invoice fraud detection and verification.

Data Science Intern

Jan 2022 – Apr 2022

Natural Resources Canada | <https://www.github.com/sharmabhay/modis250>

Ottawa, ON

- Developed **Random Forest** and **TensorFlow/Keras** models achieving **83% accuracy** and a real-time global vegetation mapping algorithm for 500m-resolution data predictions using **Google Earth Engine's JavaScript API**.

Software Developer Intern

May 2021 – Aug 2021

Ukko Agro

Toronto, ON

- Reduced API response times by **15%** by enhancing containerized **Flask** REST API middleware and **SQL** services, expanding user database and computing additional data points for users to access.

PROJECTS

GenAI Healthcare Chatbot | <https://www.github.com/sharmabhay/llm-rag-chatbot>

May 2025 – Present

- Developed a stateful, decision-based RAG chatbot using **LangGraph**, **Chroma**, and **Ollama** to host local AI models, enabling secure, conversational Q&A based on user-uploaded healthcare documents via a **Streamlit** interface.

EDUCATION

University of Waterloo

Sep 2020 – Aug 2025

- Bachelor of Mathematics (B.Math) Honours – Statistics Major, Computer Science & Economics Minor